INF1002
Programming Fundamentals
Lecture 4: Functions

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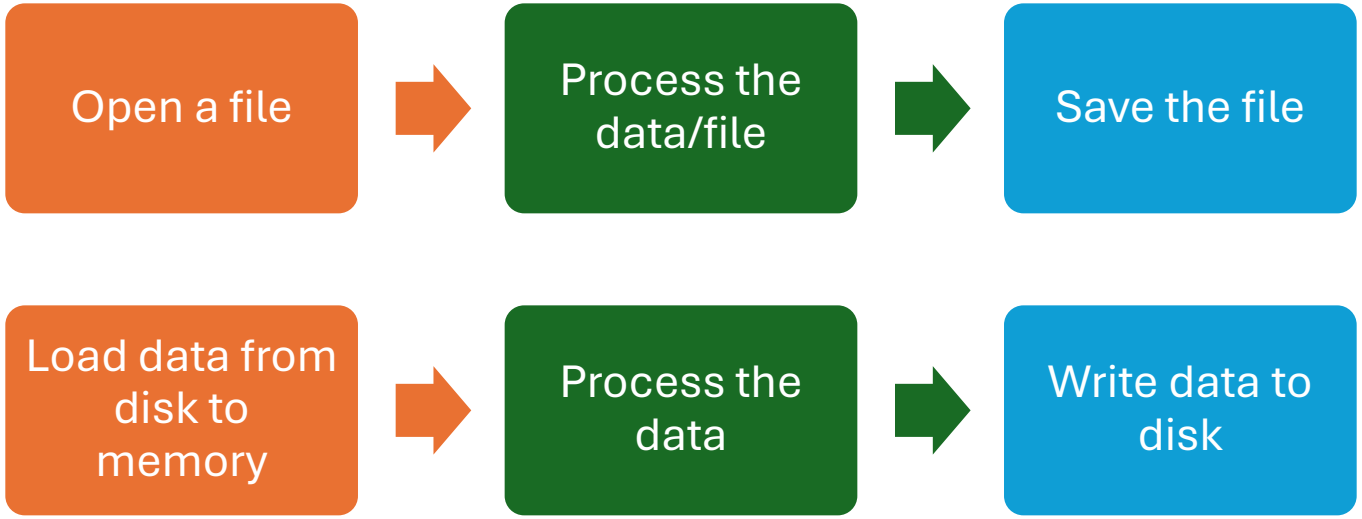
Outline

- File I/O
  - read
  - write
- Function
  - Default arguments
  - Positional and keyword arguments
  - Will a Variable's Value Change After a Function Call?
  - Variable scope
- Modules
- Higher-order functions

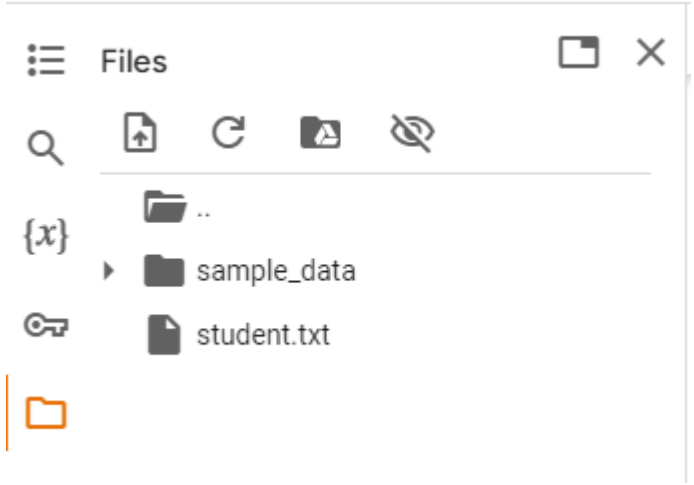
Files I/O

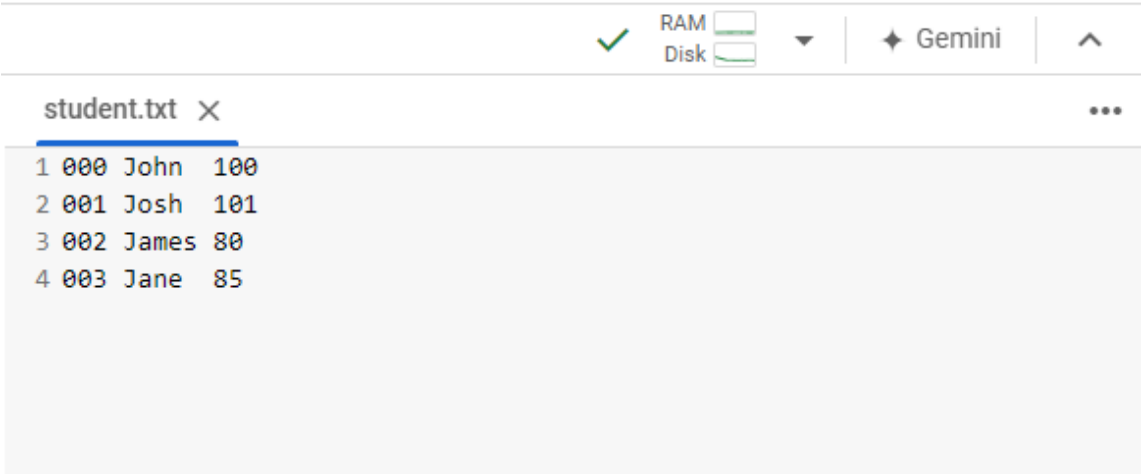
- Why do we need a file?
  - Data used in the program is stored in the memory
  - Still can find the data after reboot the computer/restart the program
  - Keep the data permanent
- File operations



Practice

- Create a new file on Google Colab
- Double click 'student.txt' and edit it, and save it (Ctrl + S)





%cd /content/drive/MyDrive

Sample file

000 John 100

001 Josh 101

002 James 89

003 Jane 95

Practice

1. Append a new student: Jack 99
2. The id of the new student is **maximum_id + 1**: 004
3. Save data to the file

```
# read file
# get max id
# generate new id
# append one student
# write to file
```

Practice

```
# read file
students = []
file = open('student.txt', 'r')
for line in file:
    line = line.strip()
    if line:
        students.append(line)
print(students)
file.close()
```

Open a file

- Python's built-in open() function
- Create one file object that can be utilized
- Syntax:
  - `file = open('student.txt', 'r')`
  - file_object= open(file_name[, access_mode] [, encoding])

open()

- `open('student.txt', 'r')`
- File_name
  - a string value that contains the name of the file that you want to access
- Access_mode
  - the mode in which the file to be opened, i.e., read, write, append, etc.
  - Optional, the default model is read(r)
- Encoding
  - Default value depends on your operating system
  - utf-8

mode

- Open('student_info.txt', mode='r')

Mode	Description
r	Opens a file for reading only. This is the default mode.
r+	Opens a file for both reading and writing.
w	Opens a file for writing only. Overwrites the file if the file exists. If the file does not exist, creates a new file for writing.
w+	Opens a file for both writing and reading. Overwrites the existing file if the file exists. If the file does not exist, creates a new file for reading and writing.
a	Opens a file for appending. If the file does not exist, it creates a new file for writing.
b	rb, wb, ab, etc. Opens the file in binary mode.
…	

readline() vs for line in file

```
while True:
    line = file.readline()  # Read one line
    if not line:  # If an empty string is returned, end of file is reached
        break
    if line.strip():
        students.append(line.strip())  # Store the line after removing the newline character
#==============================================
for line in file:
    line = line.strip()
    if line:
        students.append(line)
```

Practice

student.txt

```
000 John 100

001 Josh 101
002 James 89
003 Jane 95
```

```
#students = file.readlines()
students = [line.strip() for line in file.readlines() if line.strip()]
```

Read a file

- **readline(*size*)**
  - the *size* parameter specifies the maximum number of characters to read from the line, **strictly limiting the read to that number or until the end of the line is reached.**
- **readlines(*hint*)**
  - returns a list containing each line in the file as a list item.
  - the *hint* parameter suggests the approximate number of characters to read, but it doesn't strictly limit the total number, **ensuring complete lines are still returned**.
- read(size)

```
students = []
file = open('student.txt', 'r')
line = file.readline(3)
students.append(line)
print(students)
file.close()
exit()
```

```
students = []
file = open('student.txt', 'r')
students = file.readlines(3)
print(students)
file.close()
```

```python
# get max id
max_id = 0
for student in students:
    id = int(student.split(' ')[0])
    if id > max_id:
        max_id = id
print(max_id)
```

```python
# generate new id
new_id = max_id + 1
print(new_id)
# append a student 004
new_id = str(new_id).zfill(3)
students.append(f'{new_id} Jack 99')
print(students)
```

```python
# generate new id
new_id = max_id + 1
print(new_id)
# append a student 004
new_id = str(new_id).zfill(3)
students.append(f'{new_id} Jack 99')
print(students)
```

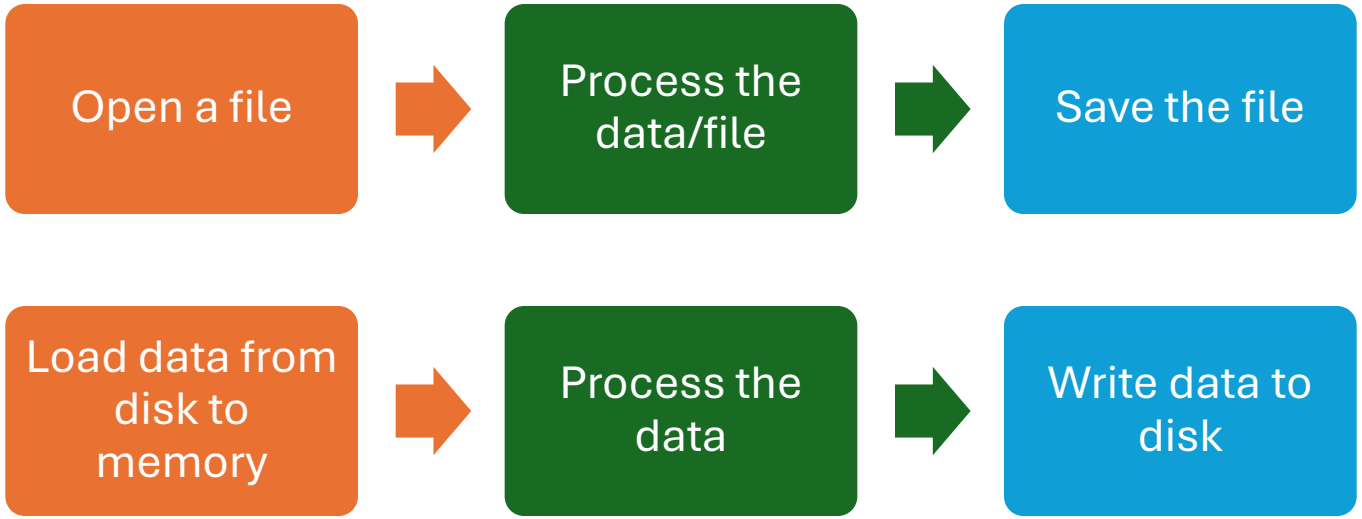
```python
num = 4
print(f"{num:03d}")
```

```
004
```

```python
# write to file
with open('student.txt', 'w', encoding='utf-8') as file:
    for student in students:
        file.write(student + '\n')
```

Write data to a file

- write(str)
- writelines(sequence)
  - Whether the lines include newline characters.



Append mode

- Open('student_info.txt', mode='r')

Mode	Description
r	Opens a file for reading only. This is the default mode.
r+	Opens a file for both reading and writing.
w	Opens a file for writing only. Overwrites the file if the file exists. If the file does not exist, creates a new file for writing.
w+	Opens a file for both writing and reading. Overwrites the existing file if the file exists. If the file does not exist, creates a new file for reading and writing.
a	**Opens a file for appending. If the file does not exist, it creates a new file for writing.**
b	rb, wb, ab, etc. Opens the file in binary mode.
…	

File pointer

- Indicate the current focused position of the file
  - open(xxx, mode='r')
  - The file pointer is placed at the beginning of the file.

```
# read file
file = open('student.txt', 'r')
seg1 = file.read(4)
print(f'{seg1=}')
seg2 = file.read(4)
print(f'{seg2=}')
file.close()
```

with open() as file:

- Automatically handles file opening and closing
- Ensures the file is properly closed even if an exception occurs

```
file = open('student.txt', 'r')
file.close()

with open('student.txt', 'r') as file:
with open('input.txt', 'r') as infile, open('output.txt', 'w') as outfile:
```

Review

- Read data from a file
- Operations on the data
- Write to a file
- File Pointer
- 'with' statement

Outline

- File I/O
  - read
  - write
- **Function**
  - Default arguments
  - Positional and keyword arguments
  - Will a Variable's Value Change After a Function Call?
  - Variable scope
- Modules
- Higher-order functions

Function

- A Python function is a block of organized, reusable code that is used to perform a single, related action.
  - Functions of a smartphone
    - Make a phone call
    - Calculator
    - Play games

https://hub.jhu.edu/2016/03/07/lego-blocks-build-better-thinkers/

https://www.tutorialspoint.com/python/python_functions.htm

function definition

```
def function_name( arguments ):
    '''function_docstring'''
    function_suite
    return [expression]
```

- Function_name
  - The identifier of the function
- Arguents
  - Input
  - Formal Arguments
  - Actual Arguments
- Function_docstring
  - Comment your codes
  - Optional
- Function_suite
  - Block of code
  - indented
- Return value
  - Optional
  - Exits a function
  - Passing back an expression to the caller
  - Output

```
import sys
def is_number(s):
    try:
        float(s)
        return True
    except ValueError:
        #print('float s error')
        return False

# write your code here
# you can use sys.argv[1] to get the first input argument.
# sys.argv[2] is the second argument, etc.
def AverageCalculator():
    input1 = sys.argv[1]
    input2 = sys.argv[2]
    input3 = sys.argv[3]
    #print(input1,input2,input3)

    if not is_number(input1) or not is_number(input2) or not
is_number(input3):
        print('Your input is invalid!')
        return
    average = (float(input1)+float(input2)+float(input3))/3
    #print(average)
    print(f'Average:{average:.2f}')
```

Arguments

- A function can have 0 or multiple arguments
- Formal Arguments
  - A place to get input value
  - A name used inside a function
- Actual Arguments
  - True value
- When you call one function, it must have the same number of arguments as defined

```python
def greetings():
    print("Hello, world!")

greetings()
```

```
Hello, world!
```

```python
def more_homework(assignments, multiplier=2, maximum_num=20):
    """
    Multiplies the number of assignments by a given multiplier.
    If the result exceeds a specified maximum, it is set to the maximum value.
```

Formal Argument

```python
# Example usage:
before = 5
after = more_homework(assignments=before, multiplier=5, maximum_num=15)
print(f'{before=},{after=}')  # Output: 15
```

Actual Argument

Positional and Keyword Arguments

- Positional Arguments
  - Passed to the function in the order in which they are defined.
  - Order matters, arguments are matched by position.
- Keyword Arguments
  - Passed to the function by explicitly naming each parameter.
  - Order doesn't matter, parameters are matched by name.

```
file = open('student.txt', 'r','utf-8')
file = open('student.txt', 'r', encoding='utf-8')
```

Positional and Keyword Arguments*

- Positional-Only Arguments
- Keyword-Only Arguments
  - To enhance the **robustness** and **readability** of the program
  - Especially when there are many parameters

Positional and Keyword Arguments*

- *args
  - Allows a function to accept any number of **positional arguments**.
  - Pack positional arguments into a tuple.

```
def sum_numbers(*args):
    total = 0
    for number in args:
        total += number
    return total

# Call the function and pass two positional arguments
result = sum_numbers(10, 20)
print(result)  # Output: 30
result = sum_numbers(1, 2, 3, 4, 5)
print(result)  # Output: 15
```

- **kwargs
  - Allows a function to accept any number of **keyword arguments**.
  - Pack keyword arguments into a dictionary.

https://www.tutorialspoint.com/python/python_arbitrary_arguments.htm

Will a Variable's Value Change After a Function Call?

- Immutable variables
  - str, int, float, tuple
  - Immutable variables are not changed after a function call

```
def AverageModifier(average):
    print(f'==inside the function {average}, id {id(average)}')
    average += 10
    print(f'==inside the function {average}, id {id(average)}')

average = 0.0
print(f'before {average}, id {id(average)}')
AverageModifier(average)
print(f'after {average}, id {id(average)}')
```

before 0.0, id 2197420871600
==inside the function 0.0, id 2197420871600
==inside the function 10.0, id 2197420578224
after 0.0, id 2197420871600

Will a Variable's Value Change After a Function Call?

- Mutable variables
  - list, dictionary
- Mutable variables can be changed as a result of one function call

```
def AddAScore(scores):
    print(f'==inside the function {scores}, id {id(scores)}')
    scores.append(10)
    print(f'==inside the function {scores}, id {id(scores)}')

scores = [0.0]
print(f'before {scores}, id {id(scores)}')
AddAScore(scores)
print(f'after {scores}, id {id(scores)}')
```

before [0.0], id 2197429867136
==inside the function [0.0], id 2197429867136
==inside the function [0.0, 10], id 2197429867136
after [0.0, 10], id 2197429867136

The scope of the variable

- Local Variable
  - Variables defined within a function can only be accessed and used inside that function.
  - They "come to life" when the function is called and "die" (or go out of scope) when the function execution is completed.

```
def add(x,y):
  z = x+y
  return z

a,b = 1,2
add(a,b)
print(z)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-18-93de7b7ed74c> in <cell line: 7>()
      5 a,b = 1,2
      6 add(a,b)
----> 7 print(z)

NameError: name 'z' is not defined
```

Next steps: Explain error

The scope of the variable

- Global Variable
- Nonlocal Variables*

```
#global variables
name = 'TutorialsPoint'
marks = 50
def myfunction():
    # accessing inside the function
    print("name:", name)
    print("marks:", marks)
# function call
myfunction()
```

```
# this is a global variable
marks = 50
def myfunction():
    # global marks
    print (marks)
    marks = marks + 20
    print (marks)

myfunction()
```

UnboundLocalError: local variable 'marks' referenced before assignment

https://www.tutorialspoint.com/python/python_variables_scope.htm

The scope of the variable

```python
marks = 50  # this is a global variable

def myfunction():
    global marks  # declare that we are referring to the global variable "marks"
    print(marks)
    marks = marks + 20
    print(marks)

myfunction()
```

```
50
70
```

```python
# this is a global variable
marks = 50
def myfunction():
    print (marks)
    local_marks = marks + 20
    print (local_marks)
myfunction()
```

```
50
70
```

Function Annotation

```python
def greet(name: str, age: int) -> str:
    return f"My name is {name}, I am {age} years old."
greet('hi',18)
```

```
'My name is hi, I am 18 years old.'
```

- Function Definition with Type Hints
- Add type annotations to function parameters and return values
- Improve readability, IDE support, and allow type checking
- Do not affect the logic of the function

Review

```
def function_name( arguments ):
    statements
```

- Arguments
  - Default arguments
  - Positional and Keyword Arguments
- Immutable variables' value cannot be changed
- Scope of variables
  - Local Variable
  - Global Variable

Outline

- File I/O
  - read
  - write
- Function
  - Default arguments
  - Positional and keyword arguments
  - Will a Variable's Value Change After a Function Call?
  - Variable scope
- **Modules**
- Higher-order functions

Modules

- Function is a container of bit of codes
- Module
  - Put related functions together.
  - A module is a file containing definition of functions, classes, variables, constants or any other Python object.
  - Python has the import keyword to load a module.

Create your own module

- A module is a simple python file
  - Like your regular scripts
- Create one module
  - Write one or more functions in a text file
  - Save it with a .py extension
  - Save in the same directory as other scripts
  - Give one descriptive name

```
'''
A module that processing text files
'''
def load_data(file_name):
    '''
    load txt lines from a file
    '''
    with open(file_name,'r') as file:
        data = file.readlines()
        return data

def process_data(data):
    '''
    data is a list of string
    for each string, there is a \\n at the end
    '''
    data = [i.strip()+': processed\n' for i in data]
    return data
    #for i in range(len(data)):
    #    data[i] = data[i].strip()+': processed\n'
```

Use a module

- import module_name [as simple_name]
- Without .py extension
- Use a function
  - module_name.fun_name()

```
import txt_processor
help(txt_processor)
#from txt_processor import load_data, process_data

file_name = 'student.txt'
data = txt_processor.load_data(file_name)
#data = load_data(file_name)
print(data)
processed_data = txt_processor.process_data(data)
#processed_data = process_data(data)
print(processed_data)
print(dir(txt_processor))
```

```
Help on module txt_processor:

NAME
   txt_processor - A module that processing text files

FUNCTIONS
   load_data(file_name)
      load txt lines from a file

   process_data(data)
      data is a list of string
      for each string, there is a \n at the end

FILE
   ---\txt_processor.py


['John\n', 'Josh\n', 'James\n', 'Jane']
['John: processed\n', 'Josh: processed\n', 'James: processed\n', 'Jane: processed\n']
['__builtins__', '__cached__', '__doc__', '__file__', '__loader__', '__name__', '__package__', '__spec__',
'load_data', 'process_data']
```

Use a module

- from module_name import (* or function_names)
  - `from txt_processor import load_data, process_data`
  - Allow you to select specific functions from the module
  - When to use from keyword
    - For a module that hundreds of functions
    - It saves loading time
- In your program, can directly use the function name

```
#import txt_processor
#help(txt_processor)
from txt_processor import load_data, process_data

file_name = 'student.txt'
#data = txt_processor.load_data(file_name)
data = load_data(file_name)
print(data)
#processed_data = txt_processor.process_data(data)
processed_data = process_data(data)
print(processed_data)
#print(dir(txt_processor))
```

Use a module

ction_names)

ns from the module

ctions

e function name

```
r

mport load_data, process_data

.txt'
r.load_data(file_name)
e_name)

t_processor.process_data(data)
processed_data = process_data(data)
print(processed_data)
#print(dir(txt_processor))
```

Python built-in modules

- Plenty of built in modules – Python really excels
- Popular modules
  - math
  - datetime
  - random
  - os
  - urllib2

```python
import math
print(math.pi)
print(math.e)
print(math.sqrt(2))
print(math.sin(math.pi/2))
print(math.cos(math.pi/2))
print(math.tan(math.pi/2))
print(math.log(1))
```

```
3.141592653589793
2.718281828459045
1.4142135623730951
1.0
6.123233995736766e-17
1.633123935319537e+16
0.0
```

File Path

- os.path.join()
- os.path.split()
- os.path.splitext()
- os.path.exists()
- os.mkdir()

Library, Package, Module

```
Library
│
└── Package
    ├── Module (module1.py)
    ├── Module (module2.py)
    └── Subpackage
        ├── Module (module3.py)
        └── Module (module4.py)
```

Popular Python Libraries for Data Science Data Mining

- Scrapy
  - Help to build crawling programs that can retrieve structured data from web
  - A great tool for scraping data used in like ML models
- BeautifulSoup
  - Another library for web crawling and data scraping
  - If you want to collect data that's available on some website but not via a proper CSV or API, this library helps you scrape it and arrange it into the format you need
- NumPy
  - A perfect tool for scientific computing and performing basic and advanced array operations
  - Handy features performing operations on n-arrays and matrics
  - Performs math operations on array
- SciPy
  - Include modules for linear algebra, integration, optimization and statistics
  - Its main functionality was built upon NumPy
  - Works great for all kinds of scientific programming projects (science, mathematics and engineering)

Popular Python Libraries for Data Science Data Mining

- Pandas
  - Help work with “labeled” and “relational” data intuitively
  - Based on two main data structures
    - Series : one-dimensional like a list of items
    - Data frames: two-dimensional like a table with multiple columns
  - A Must-have for data wrangling, manipulation and visualization
- Matplotlib
  - Standard library for visualization like two-dimensional diagrams and graphs (histograms, scatterplots, non-cartesion coordates graphs)
  - Provides an object-oriented API for embedding plots into applications
- Seaborn
  - Based on Matplotlib as useful ML tool for visualizing statistical models –heatmaps and other types of visualizations that summarize data and depict the overall distributions
  - Get to benefit from an extensive gallery of visualization including complex ones like time series, joint plots and violin diagrams

Popular Python Libraries for Data Science Data Mining

- PyTorch
  - Developed by Facebook's AI Research lab (FAIR)
  - Deep Neural Networks: Built on a tape-based autograd system
  - Flexibility: Suitable for research and prototyping
  - Performance: High performance with easy integration into existing workflows
  - Community: Strong support and extensive documentation
- Transformers
  - Developed by: Hugging Face
  - Pre-trained Models: Access to hundreds of models in over 100 languages (e.g., BERT, GPT-2, T5, RoBERTa)
  - Easy Integration: Compatible with PyTorch and TensorFlow
  - Pipeline API: Simplifies application of models to tasks like text classification, named entity recognition, question answering, and text generation
  - Makes advanced NLP models accessible and easy to deploy

Popular Python Libraries for Data Science Data Mining

- SciKit-Learn
  - An industry-standard for data science project
  - A group of packages in the SciPy stack that were created for specific functionalities like image processing
  - Uses math operations of SciPy to explosea concise interface to ML algorithms
  - HanldesML and data mining tasks such as Clustering, regression, model selection, dimensionality reduction and classification
- TensorFlow
  - A framework for ML and DL developed at Good Brain
  - Best tool for tasks like object identification, speech recognition and many others
  - Works with artificial neural networks to handle multiple datasets

Review

- import a_module as module
- Benefits of Modules
  - Code organization: Group related functionality together for better structure.
  - Code reuse: Write once, use in multiple programs.
  - Namespace isolation: Prevent naming conflicts across different parts of the project.
  - Maintainability: Easier to update and modify specific parts of the code.
- Further Reading
  - __init__.py
  - namespace

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- File I/O
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- Function
  - Default arguments
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  - Will a Variable's Value Change After a Function Call?
  - Variable scope
- Modules
- **Higher-order functions**

Functions are Objects in Python

```python
def greet(name: str, age: int) -> str:
    return f"My name is {name}, I am {age} years old."

obj1 = 'string'
obj2 = greet
print(type(obj1))
print(type(obj2))

res = obj2('abc',11)
print(res)
```

```
<class 'str'>
<class 'function'>
My name is abc, I am 11 years old.
```

- In Python, a function is an object just like a string
- The only special thing is that it’s callable.

Higher Order Functions

- Functions that take a function as an argument or return a function
  - A function can be assigned as the value of a variable
  - Can also be passed and returned just like any other reference variables
  - A high order functions can be stored in the form of lists, hash tables, etc.

```
def my_pointless(operator,number):
  return operator(number)

print(my_pointless(abs,-10))
```

```
10
```

Functions can be returned

```
def operation_factory(operation):
    def add(a, b):
        return a + b

    def subtract(a, b):
        return a - b

    if operation == 'add':
        return add
    elif operation == 'subtract':
        return subtract

add_fn = operation_factory('add')
subtract_fn = operation_factory('subtract')

print(add_fn(10, 5))
print(subtract_fn(10, 5))
```

Built-in higher order functions

- sorted()
  - Sorting items such as numbers or strings

```
scores = [70,63,98,85]
print(scores)
sorted_scores = sorted(scores)
print(sorted_scores)
```

```
[70, 63, 98, 85]
[63, 70, 85, 98]
```

  - Sorting algorithms*

sorted – 'higher order'

- Sort a list of numbers by absolute value instead by their nature value
  - Customize how sorting works
  - The sorted function takes a parameter named 'key'
  - The key needs to be a simple function that takes a single value and tells python the value to use in sorting it
  - Built-in function abs() can be used to get the absolute value
  - We can specify the abs() as the key argument

```python
scores = [70,-63,-98,85]
print(scores)
sorted_scores = sorted(scores)
print(sorted_scores)
sorted_scores = sorted(scores, key=abs)
print(sorted_scores)
```

```
[70, -63, -98, 85]
[-98, -63, 70, 85]
[-63, 70, 85, -98]
```

sorted – 'higher order'

- Specify your own function to sort the list
- Example:
  - Given a list of strings, sort based on their last character
  - Write one function to get the last character of the one string
  - Specify this function as the key of the sorted function

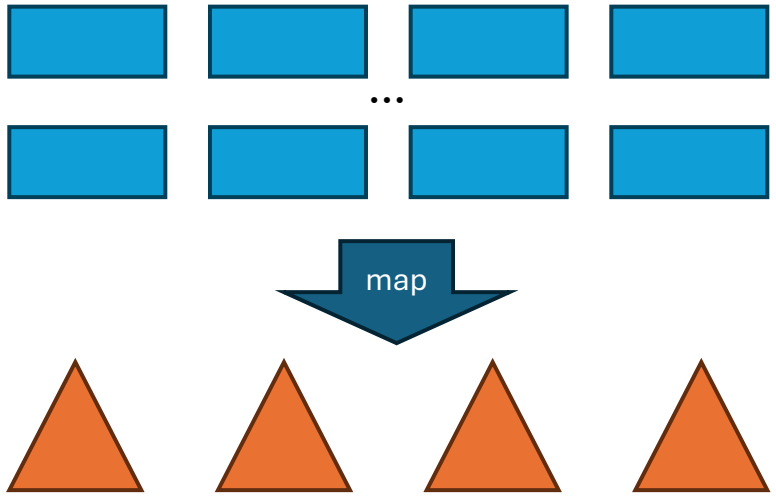
```python
def last_char(s):
  return s[-1]

names = ['JohnB','JaneA','JoshC','JamesD']
sorted_names = sorted(names)
print(sorted_names)
sorted_names = sorted(names, key=last_char)
print(sorted_names)
```

```
['JamesD', 'JaneA', 'JohnB', 'JoshC']
['JaneA', 'JohnB', 'JoshC', 'JamesD']
```

Built-in higher order functions – map

- **map**(*function*, *iterable*, **iterables*)
- map(func, sequence(s))
  - One of the arguments is a function
  - The function is applied to each item in the sequence
  - Return a sequence with different values
  - Same order, same length, mapped via a function



Definition of iterable*

iterable

An object capable of returning its members one at a time. Examples of iterables include all sequence types (such as `list`, `str`, and `tuple`) and some non-sequence types like `dict`, file objects, and objects of any classes you define with an `__iter__()` method or with a `__getitem__()` method that implements sequence semantics.

Iterables can be used in a `for` loop and in many other places where a sequence is needed (`zip()`, `map()`, ...). When an iterable object is passed as an argument to the built-in function `iter()`, it returns an iterator for the object. This iterator is good for one pass over the set of values. When using iterables, it is usually not necessary to call `iter()` or deal with iterator objects yourself. The `for` statement does that automatically for you, creating a temporary unnamed variable to hold the iterator for the duration of the loop. See also iterator, sequence, and generator.

https://docs.python.org/3/glossary.html

map

- Example
  - Given a list of numbers, calculate the squares of each item
  - Define one function square(x) to calculate the square number
  - Use the map function to calculate the squares of the list

```python
def square(num):
  return num**2

numbers = [1,2,3,4,5]
squares = map(square, numbers)
print(squares)
print(list(squares))
```

```
<map object at 0x7cd9e7ac23e0>
[1, 4, 9, 16, 25]
```

```python
my_squares = []
for num in numbers:
  my_squares.append(square(num))
print(my_squares)
```

map

- map can also use built-in functions like abs
- Practice
  - Suppose you have a list of student names and scores
  - ['John\t100', 'Josh\t90','Jane\t101','James\t92']
  - Get the scores only

Map - multiple iterables

- The number of parameters in the function must match the number of iterables.
- The output iterator will have the same length as the shortest iterable.
- Each item in the result is computed by taking one element from each iterable in parallel and passing them together into the function.

```python
nums1 = [1, 2, 3, 4]
nums2 = [10, 20, 30]
def add(x,y):
  return x+y

result = map(add, nums1, nums2)
print(list(result))
```

```
[11, 22, 33]
```

Return value of map: Iterator

- In Python, an iterator is an object that lets you loop over a sequence of values, one at a time.
- You can call next() on to get values one by one.
- When there are no more values, it raises StopIteration.

```python
nums = [1, 2, 3]
squares = map(lambda x: x*x, nums)

print(type(squares))        # class 'map'
print(next(squares))  # 1
print(next(squares))  # 4
print(next(squares))  # 9
print(next(squares))  # StopIteration
```

```
<class 'map'>
1
4
9
---------------------------------------------------------------------------
StopIteration                             Traceback (most recent call last)
/tmp/ipython-input-1777475477.py in <cell line: 0>()
      6 print(next(squares))  # 4
      7 print(next(squares))  # 9
----> 8 print(next(squares))  # StopIteration

StopIteration:
```

Next steps: Explain error

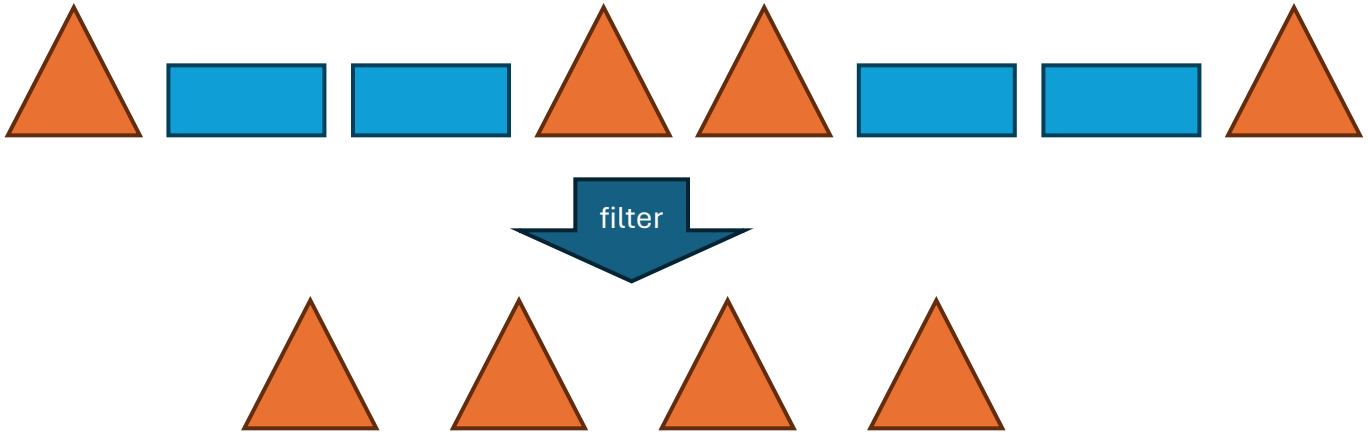
Return value of map: Iterator

- Lazy (on-demand calculation)
  - Values are produced only when needed
  - Saves memory (doesn’t store all results at once)

- One-time use (forward only)
  - Use next() to get values one by one
  - No previous(), no indexing
  - Once consumed, cannot restart or go back

Built-in higher order functions - filter

- filter (func, sequence)
  - func: the filtering function
  - seq: a sequence of the values
  - Output: a sequence of values for which the function func returns true
- Function applies to each item to decide whether the item shall be kept or not
  - If function returns True, the item is kept
  - Otherwise, it is removed



filter

- Given a list of strings, filter those are start with 't'
  - Define a function names is_startwitht(string) as the filter function
  - Do the filter

```
def is_startwitht(string):
  return string.startswith('t')

items = ['test','r','s','trangle','tr','st','bo']
t_items = filter(is_startwitht, items)
print(list(t_items))
```

```
['test', 'trangle', 'tr']
```

  - Implement the same function using for loop

Map, Filter, and List Comprehension

```python
def double(x):
    return x * 2

numbers = [1, 2, 3]
result = map(double, numbers)
print(list(result))  # Output: [2, 4, 6]

result = [double(x) for x in numbers]
print(result)  # Output: [2, 4, 6]
```

```python
def is_even(x):
    return x % 2 == 0

numbers = [1, 2, 3, 4]
result = filter(is_even, numbers)
print(list(result))  # Output: [2, 4]

result = [x for x in numbers if is_even(x)]
print(result)  # Output: [2, 4]
```

Map, Filter, and List Comprehension

- Lazy evaluation
  - map() and filter() return iterators that delay computation until results are requested (e.g., with list(), for loop, or next()).
  - Expressions are evaluated only when needed.
- Eager evaluation
  - List comprehension immediately calculates and returns a complete list.

reduce

- reduce (func, seq)
  - Takes a sequence of data and return a single value
  - Func: Takes two values and returns one value
    1. Consume the first two values from the sequence
    2. Return a value
    3. Consume the next value and the returned value
    4. Repeat 3, until all values in the sequence have been consumed
    5. Return the final value
- In python3, it is not a built-in function any longer. It is moved to 'functools' module
- from functools import reduce

reduce

- Calculate the summation of all the numbers in one list using reduce
- Define a function that adds two numbers
- reduce (func, seq)

```
from functools import reduce

numbers = [1,2,3,4,5,6]

def add(x,y):
  return x+y

the_sum = reduce(add, numbers)
print(the_sum)
```

```
21
```

Notes

- All these functions does not change the original sequence
- A new sequence is created to hold the output

Review

- Sorted
- Map and Filter
  - Lazy evaluation
- Reduce